

Where Does Adversarial Vulnerability Come From? A Controlled Gaussian Study with a Real-Data Check

Abstract

Why are accurate classifiers so easy to fool with tiny perturbations? We study the smallest setting in which the question has an exact answer: Gaussian inputs whose Bayes-optimal classifier is linear. There, the ℓ_∞ flip radius is $|\text{LLR}|/\|w\|_1$ with $\|w\|_1 = \sum_i d'_i/\sigma_i$, and five synthetic experiments show that this quantity, not any single fragile feature, explains what trained networks do. Vulnerability grows when (i) the same evidence is *spread* over more dimensions (radius $\propto k^{-1/2}$ in ℓ_∞ , unchanged in ℓ_2), and (ii) a feature’s *scale* (its class-mean gap) is small next to the budget, which is Ilyas et al.’s notion of non-robustness. It does *not* grow with a feature’s statistical reliability: at fixed scale, a $4\times$ more reliable feature is no safer. A trained MLP matches the Bayes-optimal classifier’s vulnerability, so this is a property of the data and threat model, not of training; no single weak dimension is fragile, only their sum; and interventions that help do so by shrinking $\|w\|_1$ and pay in accuracy. A mixing test on MNIST (E6) confirms the geometry controls but not the $\sqrt{k}$ law: natural pixels are already near the saturated regime, and on CIFAR-10 the ℓ_∞ cost of using more directions comes from a shrinking ℓ_2 margin, not from spread. We state what would falsify the account, where it is weakest, and where we expect readers to disagree.

1 The question and the claim

Ilyas et al. [3] propose that adversarial examples exploit *non-robust features*: predictive, but brittle. Goodfellow et al. [2] proposed earlier that the cause is linearity in high dimension: a perturbation of size ε in every coordinate moves a linear score by $\varepsilon\|w\|_1$, which grows with dimension even when each coordinate’s change is imperceptible. These stories are usually told as alternatives. We ask what a fully controlled example says about them.

Claim. In a task where the optimal classifier is linear, ℓ_∞ vulnerability is set by $\|w\|_1 = \sum_i d'_i/\sigma_i$: the evidence the classifier collects per unit of absolute noise scale, summed over dimensions. Two things make it large: evidence *spread* over many dimensions, and features of small *scale* (class-mean gap $\Delta\mu_i$ small relative to the budget ε), the operational meaning of “non-robust” in [3]. A feature’s statistical *reliability* d' is not a cause. The trained network inherits all of this from the data; it is not a learning failure.

What would falsify it. The claim predicts five things, each of which could have come out otherwise:

- P1** At fixed information, the minimal ℓ_∞ flip radius falls as $k^{-1/2}$ in the number k of dimensions sharing it, while the ℓ_2 radius is unchanged.
- P2** A trained MLP is no more vulnerable than the Bayes-optimal classifier. If it were markedly worse, there would be learned fragility beyond what the data imposes.

- P3** Each weak dimension, attacked alone, is *harder* to flip than the strong one. Only their sum is dangerous.
- P4** Interventions that raise robustness do so by lowering $\|w\|_1$ and pay for it in accuracy; the exchange rate is set by the discarded information.
- P5** For a single feature, the flip radius is proportional to its scale $\Delta\mu$ and nearly independent of its reliability d' at fixed $\Delta\mu$. If reliability were the cause, a more reliable feature would be safer.

Terminology. Two words are kept apart throughout. A dimension is *weak* if it is statistically unreliable on its own (small d'). It is *non-robust at budget ε* if its class-conditional correlation does not survive the worst-case ε -perturbation, which for a Gaussian feature means $\Delta\mu_i/2 < \varepsilon$ (Ilyas et al.’s operational definition, applied to our task). Both are fixed by the generative model *before* any attack is run, which avoids the circularity of calling “non-robust” whatever the attack succeeded through. In the task of Table 1 the weak dimensions are also non-robust at $\varepsilon = 0.3$ (half-gap $0.2 < 0.3$) and the strong one is neither, so Experiments 2–4 cannot separate weakness from scale. Experiment 5 does.

2 The mechanism in one equation

Let $y \in \{0, 1\}$ be balanced and, conditionally on y , let the inputs be independent Gaussians

$$x_i = \left(y - \frac{1}{2}\right) \Delta\mu_i + \sigma_i \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1),$$

with per-dimension discriminability $d'_i = \Delta\mu_i/\sigma_i$. The inputs are deliberately *not* clipped to $[0, 1]$: the Bayes rule is then exactly linear and the budget is not confounded with the size of the domain. The log-likelihood ratio is

$$\text{LLR}(x) = \sum_i w_i x_i, \quad w_i = \frac{\Delta\mu_i}{\sigma_i^2}, \quad \text{Bayes accuracy} = \Phi\left(\frac{1}{2} \sqrt{\sum_i d_i'^2}\right).$$

The smallest perturbation that flips a correctly classified point is, by duality of norms,

$$\varepsilon_\infty(x) = \frac{|\text{LLR}(x)|}{\|w\|_1}, \quad \varepsilon_2(x) = \frac{|\text{LLR}(x)|}{\|w\|_2}. \quad (1)$$

Now spread a fixed amount of evidence d' equally over k dimensions with the same noise level σ , so $d'_i = d'/\sqrt{k}$. Then $w_i = d'/(\sqrt{k}\sigma)$ and

$$\|w\|_2 = \frac{d'}{\sigma} \quad (\text{independent of } k), \quad \|w\|_1 = \sqrt{k} \frac{d'}{\sigma}. \quad (2)$$

The classifier’s accuracy and ℓ_2 margin are identical for every k ; its ℓ_∞ margin shrinks like $k^{-1/2}$. This is the whole mechanism: an ℓ_∞ ball lets the attacker move all k coordinates by ε in the worst-case direction, while an ℓ_2 ball of the same radius must share its budget across them. Everything below tests whether this accounts for what trained networks do.

Metric. For each correctly classified test point we compute the minimal flipping radius: exactly from (1) for the Bayes rule, and by per-point bisection over a 20-step PGD attack for trained networks (we verified that on the Bayes rule this agrees with (1) to within the bisection tolerance, 2.4×10^{-4}). The empirical CDF of this radius *is* the attack-success-versus- ε curve, so one number per point replaces a grid of budgets. Models are one-hidden-layer MLPs (8 ReLU units, Adam, 500 steps, weight decay 10^{-2}) on 4,000 training points; all results use 2,000 test points and 10 seeds, reported as mean $\pm$ standard deviation over seeds of the per-seed median.

Table 1: The task for Experiments 2–4: one strong dimension and five weak ones. Single-dimension accuracy is $\Phi(d'/2)$. Together the five weak dimensions have a larger total weight (32.0) than the strong one (11.1) and slightly more squared discriminability (12.8 vs. 11.1), yet individually each is *weak*.

Dimension	$\Delta\mu$	σ	d'	single-dim. accuracy	Bayes weight w_i
strong (x_0)	1.0	0.30	3.33	0.952	11.1
weak ($x_1 \dots x_5$, each)	0.4	0.25	1.60	0.788	6.4
all six, Bayes-optimal			4.89	0.993	$\ w\ _1 = 43.1$, $\ w\ _2 = 18.1$

3 Experiments

3.1 E1. The same evidence, spread over more dimensions

Design. Fix the total discriminability at $d' = 1/0.3$ (Bayes accuracy 95.2%) and split it equally over $k \in \{1, 2, 4, \dots, 64\}$ dimensions, each with $\sigma = 0.25$. Accuracy is the same at every k by construction. Only the *number of places the evidence lives* changes. We train an MLP at each k and compute the Bayes rule exactly.

Result. (Figure 1, Table 2.) The median ℓ_∞ radius falls from 0.43 at $k=1$ to 0.054 at $k=64$, a log-log slope of -0.4995 for the Bayes rule and -0.4987 for the MLP, against the predicted -0.5 . The median ℓ_2 radius is constant (0.43 at every k , slope 0.000 and -0.001). The trained MLP lies on the exact curve at every k , with seed-to-seed standard deviation at most 0.006.

Table 2: E1 numbers. Median minimal flip radius; MLP: mean $\pm$ s.d. over 10 seeds. Bayes values are exact, averaged over test sets.

k	$\ w\ _1/\ w\ _2$	Bayes ε_∞	MLP ε_∞	Bayes ε_2	MLP ε_2	MLP acc.
1	1.00	0.432	0.433 $\pm$ 0.006	0.432	0.433 $\pm$ 0.006	0.951 $\pm$ 0.004
4	2.00	0.214	0.215 $\pm$ 0.003	0.429	0.429 $\pm$ 0.006	0.955 $\pm$ 0.005
16	4.00	0.108	0.108 $\pm$ 0.002	0.432	0.430 $\pm$ 0.007	0.953 $\pm$ 0.004
64	8.00	0.054	0.054 $\pm$ 0.001	0.432	0.429 $\pm$ 0.007	0.950 $\pm$ 0.005

Reading. P1 holds. A classifier can lose a factor of eight in ℓ_∞ robustness, with no change in accuracy, no change in ℓ_2 robustness, and no individually fragile feature, purely because its evidence is divided among more coordinates. The ratio $\varepsilon_2/\varepsilon_\infty$ equals $\|w\|_1/\|w\|_2$ at every k .

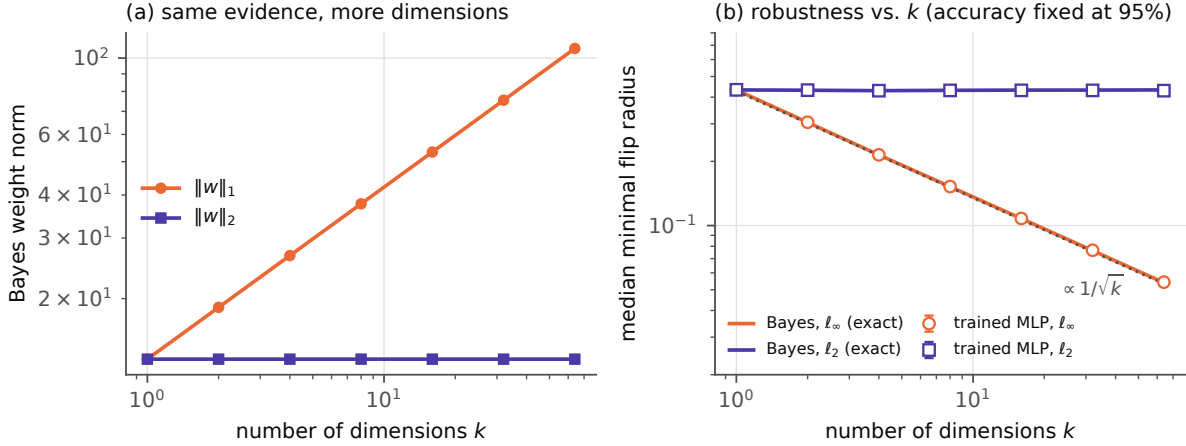


Figure 1: E1. (a) At fixed total evidence, $\|w\|_2$ is constant and $\|w\|_1$ grows as $\sqrt{k}$ (Eq. 2). (b) Median minimal flip radius versus k : exact Bayes (lines) and trained MLP (markers, ± 1 s.d. over 10 seeds, smaller than the markers for ℓ_∞). Accuracy is 95% throughout.

3.2 E2. The optimal classifier is as vulnerable as the trained one

Design. Use the task of Table 1. Attack the trained MLP and the exact Bayes-optimal classifier. The Bayes classifier has no optimisation, no overfitting and no inductive bias.

Result. (Figure 2.) The two curves nearly coincide. Median ε_∞ is 0.277 ± 0.002 (Bayes) versus 0.285 ± 0.002 (MLP); the fraction of points flipped at $\varepsilon = 0.3$ is 58.1% versus 55.3%. Clean accuracy is 0.9926 versus 0.9924 ± 0.0014 . If anything, the trained network is slightly *more* robust than the optimum.

Reading. P2 holds. For this task, the vulnerability is a property of *the data and the threat model*, not of how the network was trained. This is also why the companion note’s observation [1] that the MLP relies on the weak dimensions *less* than a Bayes-optimal classifier does is consistent with its being vulnerable: the weights the optimum places there are already enough.

3.3 E3. Who is vulnerable? No dimension is; the sum is

Design. Restrict the attacker to a subset of coordinates and measure ε_∞ for the Bayes rule and the MLP, again on Table 1’s task. This is the control missing from the usual “attack only the weak features” experiment, which compares five dimensions to one and so confounds feature type with attack dimensionality.

Table 3: E3. ℓ_∞ attack restricted to a subset S of dimensions: median minimal flip radius (larger = harder) and fraction flipped at $\varepsilon = 0.3$. For the Bayes rule $\varepsilon_\infty = |\text{LLR}|/\|w_S\|_1$ exactly.

Attacked dims S	$\ w_S\ _1$	Bayes ε_∞	MLP ε_∞	Bayes flipped	MLP flipped
strong dimension (1)	11.1	1.076	1.012 ± 0.018	3.3%	3.8%
one weak dimension	6.4	1.868	2.005 ± 0.078	1.3%	1.3%
all 5 weak dimensions	32.0	0.374	0.397 ± 0.006	31.1%	27.3%
all 6 dimensions	43.1	0.277	0.285 ± 0.002	58.1%	55.3%

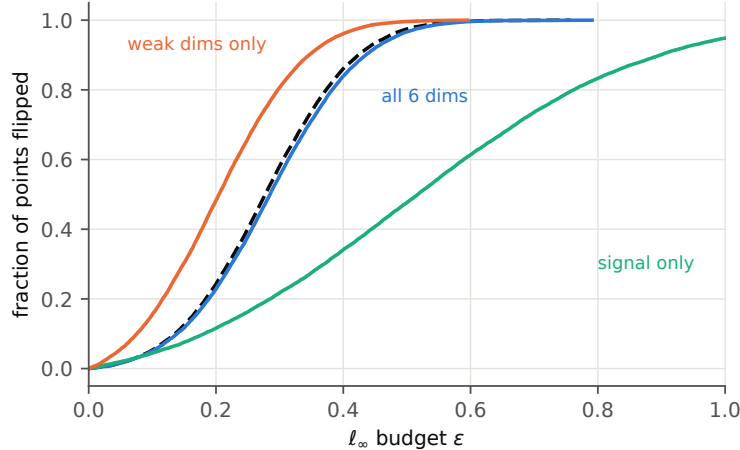


Figure 2: E2 and E4. Fraction of test points flipped by an ℓ_∞ attack of budget ε (pooled over seeds). Dashed black: exact Bayes-optimal classifier on all six dimensions; it lies on top of the trained MLP (blue). Models restricted to the weak dimensions only (orange) or the strong one only (green) are shown for comparison (E4).

Reading. P3 holds. A single weak dimension is harder to flip than the strong one (1.87 vs. 1.08): its weight is smaller. The five together are $\approx 3\times$ easier to flip than the strong one (0.37 vs. 1.08) because $\|w_S\|_1$ adds up (32.0 vs. 11.1). The vulnerability is therefore not located in any feature. This corrects the common summary that “attacks on the weak features are more effective”: per dimension they are less effective, per attacker they are more, simply because there are more of them.

3.4 E4. Interventions work by shrinking $\|w\|_1$, and pay in accuracy

Design. Compare: all six dimensions; the strong dimension only (removing the weak dimensions from the input); the five weak dimensions only (the converse); and all six with a first-layer L_1 penalty $\lambda \in \{0.02, \dots, 0.2\}$ pushing weights onto fewer dimensions. Report ℓ_∞ and ℓ_2 radii.

Table 4: E4. Clean accuracy and median minimal flip radius (mean $\pm$ s.d. over 10 seeds). The Bayes-optimal rule on the same inputs gives $\varepsilon_\infty = 0.517$ (strong only), 0.204 (weak only) and 0.277 (all six).

Model	accuracy	ε_∞	ε_2	$\ w\ _1/\ w\ _2$ (Bayes)
all six (MLP)	0.992 $\pm$ 0.001	0.285 $\pm$ 0.002	0.666 $\pm$ 0.004	2.38
strong only	0.953 $\pm$ 0.005	0.517 $\pm$ 0.006	0.517 $\pm$ 0.006	1.00
weak only (5 dims)	0.962 $\pm$ 0.003	0.205 $\pm$ 0.002	0.458 $\pm$ 0.004	2.24
all six, $L_1 \lambda = 0.05$	0.987 $\pm$ 0.002	0.343 $\pm$ 0.005	0.651 $\pm$ 0.006	—
all six, $L_1 \lambda = 0.10$	0.976 $\pm$ 0.004	0.402 $\pm$ 0.010	0.598 $\pm$ 0.011	—
all six, $L_1 \lambda = 0.20$	0.957 $\pm$ 0.005	0.496 $\pm$ 0.010	0.529 $\pm$ 0.006	—

Result. Three observations. (i) Removing the weak dimensions raises ε_∞ from 0.285 to 0.517 (1.8 $\times$) and costs 4.0 points of accuracy. (ii) The weak-only model is the instructive control: at 96.2% accuracy ($>$ strong-only) it is 2.5 $\times$ less ℓ_∞ -robust. Which dimensions you keep matters

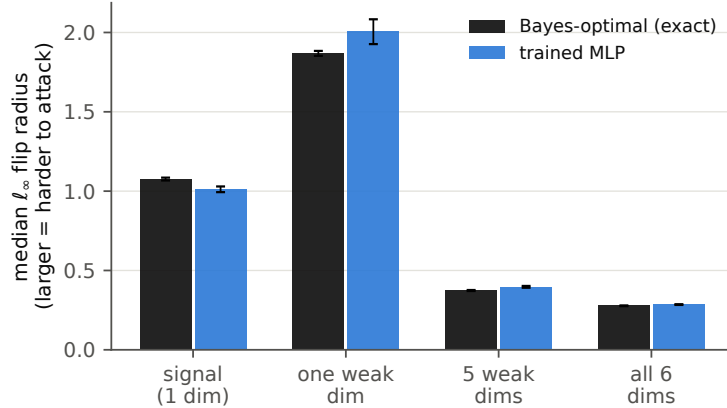


Figure 3: E3. Median ℓ_∞ flip radius when the attacker controls only the listed dimensions. One weak dimension is the *hardest* to exploit; five together are about three times easier than the strong dimension alone.

less than how many share the evidence. Its ℓ_2 radius (0.458) is close to the strong-only model’s (0.517), and $\varepsilon_2/\varepsilon_\infty = 2.23$ matches $\|w\|_1/\|w\|_2 = \sqrt{5}$ as predicted. (iii) The L_1 sweep walks close to a line from the Bayes point to the strong-only point: ℓ_∞ radius rises 74% while accuracy falls 3.6 points. Along it the ℓ_2 radius *falls* (0.666 $\rightarrow$ 0.529): the penalty concentrates weight, which reduces $\|w\|_1$ at fixed $\|w\|_2$ and trades one kind of robustness for accuracy rather than creating robustness outright.

Reading. P4 holds. Robustness gained by sparsifying is paid for with the evidence discarded; nothing in these runs offers a free lunch. Note what this does *not* show: that a smarter regulariser or architecture could not do better on real data. It shows that, here, the exchange is set by the data.

3.5 E5. Scale, not reliability: one feature, two knobs

Design. Everything so far moved weak features and strong features together. Here we pull the two apart in the simplest possible case: a single Gaussian feature with class-mean gap $\Delta\mu \in \{0.05, \dots, 0.8\}$ and reliability $d' \in \{1.5, 3, 6\}$, with $\sigma = \Delta\mu/d'$. Reliability sets accuracy (0.77, 0.93, 0.999); scale sets robustness in Ilyas et al.’s sense. Trained models carry a fixed standardisation layer so that tiny-scale inputs are not handicapped by the optimiser; attacks act on raw inputs.

Result. (Figure 5, Table 5.) The median flip radius is proportional to $\Delta\mu$ across a $16\times$ range, with a constant $\varepsilon_\infty/\Delta\mu = 0.69, 0.53, 0.50$ for $d' = 1.5, 3, 6$ (to two digits at every scale). A $4\times$ change in reliability changes the radius by $1.4\times$, in the wrong direction for a reliability explanation: the *more* reliable feature is slightly *less* robust, approaching the limit $\Delta\mu/2$ (in one dimension the flip radius is just the distance to the decision boundary). The trained MLP matches the exact Bayes radius to within 0.0011 in every cell.

Reading. P5 holds. A single, nearly perfectly reliable feature ($d' = 6$, 99.9% accuracy) of small scale is as attackable as a radius of $\Delta\mu/2$ allows: at $\Delta\mu = 0.05$ it flips at $\varepsilon = 0.025$. This is what “non-robust” means in [3], and it is a different knob from E1’s spread. Both enter the same formula, $\|w\|_1 = \sum_i d'_i/\sigma_i$: E1 grows it through the number of terms, E5 through a smaller σ_i at fixed d'_i . A caution on E1: there each dimension’s $\Delta\mu_k = d'\sigma/\sqrt{k}$ shrinks with k as well, so E1

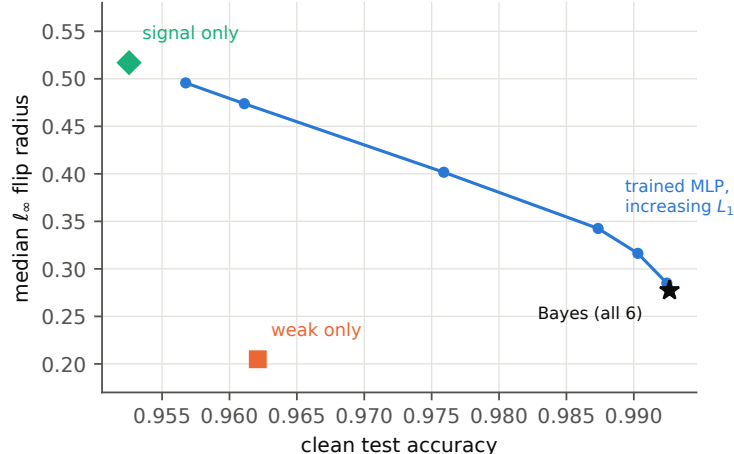


Figure 4: E4. Accuracy versus ℓ_∞ robustness. Increasing the L_1 penalty (blue) moves the model from the Bayes-optimal point (star) toward the strong-only solution (green diamond). The weak-only model (orange) is both more accurate than strong-only and far less robust.

Table 5: E5. Median ℓ_∞ flip radius (exact Bayes; MLP within 0.0011) and, in parentheses, the ratio $\varepsilon_\infty/\Delta\mu$, which depends on d' alone.

$\Delta\mu$	$d' = 1.5$ (acc. 0.77)	$d' = 3$ (acc. 0.93)	$d' = 6$ (acc. 0.999)
0.05	0.035 (0.69)	0.027 (0.53)	0.025 (0.50)
0.1	0.070 (0.70)	0.053 (0.53)	0.050 (0.50)
0.2	0.138 (0.69)	0.106 (0.53)	0.100 (0.50)
0.4	0.276 (0.69)	0.210 (0.53)	0.200 (0.50)
0.8	0.560 (0.70)	0.422 (0.53)	0.400 (0.50)

alone does not show that spread is separate from scale. What separates them is the ℓ_2 control: if per-coordinate scale were the cause, the ℓ_2 radius would move with k too, and it does not.

4 Does it survive on real data? E6

Design. We test the spread mechanism on MNIST by changing only how widely information is spread across coordinates. Inputs (centred, unclipped) are mixed by a fixed block-diagonal random orthogonal matrix Q_k with block size $k \in \{1, 4, 16, 49, 196, 784\}$ over a random pixel permutation. Q_k preserves information and ℓ_2 geometry exactly; only the coordinate basis changes. An MLP (784–256–10, SGD with momentum and ℓ_2 weight decay, which are rotation-equivariant; Adam is not) is trained and attacked in the mixed coordinates with the same minimal-radius metric (10,000 training and 1,000 test points, 5 seeds), on 3-vs-7 and on all ten classes. Predictions: accuracy and ε_2 flat in k ; ε_∞ falling as $k^{-1/2}$.

Result. The controls hold: accuracy and ε_2 do not change with k (accuracy within 0.001; ε_2 within seed noise, s.d. 0.02–0.04). The ℓ_∞ prediction *fails as stated*: ε_∞ falls by only 12% (3-vs-7) and 14% (ten classes) between $k=1$ and $k=784$, not by the factor of 28 that $k^{-1/2}$ would give. The cause is visible in the last column: $\varepsilon_2/\varepsilon_\infty$, the trained network’s analogue of $\|w\|_1/\|w\|_2$, is already 19–20 in the pixel basis and saturates at 22.4, the value for a dense isotropic direction. Equation (1)

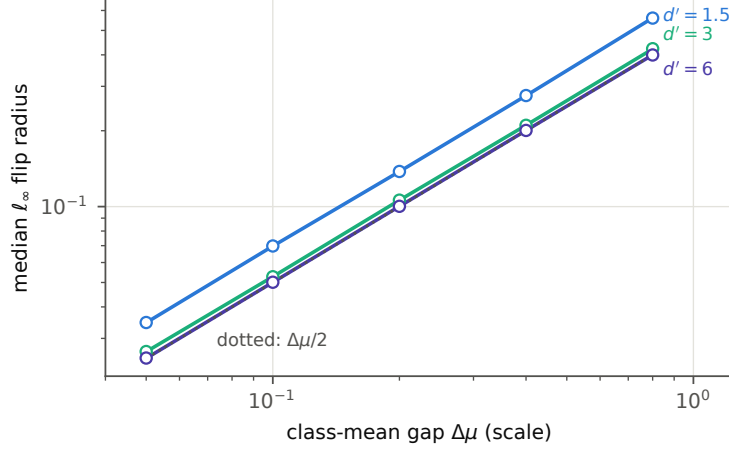


Figure 5: E5. Median ℓ_∞ flip radius of a single-feature classifier versus its class-mean gap $\Delta\mu$, at three reliabilities. Lines: exact Bayes; markers: trained MLP (± 1 s.d., 10 seeds). Radius tracks scale; reliability only shifts the line by a factor of at most 1.4.

Table 6: E6. MNIST, median minimal flip radius, mean over 5 seeds (s.d. in text). The last column is $\varepsilon_2/\varepsilon_\infty$; a dense isotropic direction gives $\sqrt{2/\pi}\sqrt{784} = 22.3$.

Task	k	accuracy	ε_∞	ε_2	$\varepsilon_2/\varepsilon_\infty$
3 vs. 7	1	0.991	0.0687	1.351	19.7
	16	0.991	0.0617	1.354	22.0
	196	0.991	0.0607	1.358	22.4
	784	0.991	0.0605	1.352	22.4
10 class	1	0.956	0.0481	0.929	19.3
	16	0.957	0.0421	0.919	21.8
	196	0.955	0.0416	0.927	22.3
	784	0.956	0.0413	0.925	22.4

still organises the data: the radius is set by the ℓ_1 -to- ℓ_2 ratio of the classifier’s gradient. But MNIST in its natural basis is already in the saturated regime, so the spread knob of E1 has little room left to turn. We read this as a limit on the claim: the $\sqrt{k}$ law describes how vulnerability grows with spread, not that natural images are far from saturation, and E6 does not show the latter either way. Mixing is a change of basis, not a new dataset.

E6b. CIFAR-10, number of directions used. A second real-data check varies how many input directions the network may use. A five-layer CNN (20,000 training and 500 test points, 5 seeds, SGD) sees only the projection of the image onto the top- m PCA directions; the attacker perturbs raw pixels with an unclipped ℓ_∞ or ℓ_2 ball, so only the in-subspace component matters.

Using more directions buys accuracy ($0.44 \rightarrow 0.78$) and costs ℓ_∞ robustness by a factor of 6.8, which is the exchange of P4 on real data. But the mechanism is not spread: $\varepsilon_2/\varepsilon_\infty$ is near its dense limit at every m and *decreases* slightly with m , while ε_2 itself falls by $7.9\times$. The ℓ_∞ loss is therefore almost entirely a shrinking ℓ_2 margin, i.e. the network relies on lower-variance, smaller-margin directions (the scale route of E5, not the spread route of E1). Caveat: medians are

Table 7: E6b. CIFAR-10, median minimal flip radius over correctly classified points (mean over 5 seeds). A dense isotropic direction gives $\varepsilon_2/\varepsilon_\infty = 44.2$.

m	accuracy	ε_∞	ε_2	$\varepsilon_2/\varepsilon_\infty$
16	0.442	0.0176	0.779	44.3
64	0.574	0.0102	0.431	42.2
256	0.704	0.0057	0.230	40.6
1024	0.780	0.0030	0.116	39.1
3072	0.778	0.0026	0.099	38.5

over correctly classified points, so low- m models are evaluated on an easier subset, which overstates their robustness.

Earlier image result. The companion note [1] reports that a background-only attack flips 67% of predictions at $\varepsilon = 0.3$ (FGSM) on a circle-vs-square task, and that a two-branch CNN reduces this to 31%. That used a different metric, has not been re-run here, and we do not rely on it.

5 Discussion: what this does and does not say

Relation to non-robust features. The answer to “do non-robust features cause vulnerability?” is yes, with two qualifications. First, non-robust must be read as Ilyas et al. define it, small class separation relative to the budget (scale, E5), and not as “statistically weak”: a nearly perfect small-scale feature is the most attackable thing in this paper. Second, the vulnerable direction in the six-dimension task is made of statistically *legitimate* features: the Bayes classifier wants them, and the five weak dimensions together have $2.9\times$ the weight $\|w_S\|_1$ of the strong one (and $1.15\times$ its squared discriminability). A coalition of small contributions is easy to move, which is the spread effect (E1, E3). We do not claim this is the whole of Ilyas et al.’s account on natural images, where non-robust features also have a frequency character that this toy does not model.

Relation to prior work. The ingredients are not new: the $\varepsilon\|w\|_1$ growth of linear scores is Goodfellow et al.’s [2] linearity argument, with ℓ_1 -gradient norms and input dimension linked in [6]; Gaussian class-conditional models with one strong and many weak features appear in [4, 5]. What is added here is the isolation: information held *fixed*, only its spread varied, with exact predictions checked against trained networks and against the Bayes rule, and a norm-swap control (ℓ_2 unchanged) that distinguishes accumulation from fragility, and a factorial (E5) that separates a feature’s scale from its reliability.

Where you may disagree.

- “*The task is linear, so of course.*” Yes, by design: it is the case where the claim can be derived rather than argued. That a one-hidden-layer MLP lands on the linear solution (E1, E2) shows the claim holds for a network here, not that deep networks on natural data share it.
- “ *ℓ_∞ is a convention, not a fact.*” Agreed, and the result says so: under ℓ_2 the same classifiers are indistinguishable across k . The vulnerability is a statement about the pairing of a data distribution with a threat model.

- “*Independent features are unrealistic.*” Correlated weak features share noise and so carry less independent evidence; that would weaken the effective k , not remove the mechanism. We have not tested it.
- “*Robust training obviously fixes this.*” It may, and adversarial training is known to. Nothing here concerns whether it *costs* what this account predicts on real data.

Limitations. One task family, equal-weight Gaussian dimensions, a single small architecture, standardised inputs for E5 (without which a tiny-scale feature is harder to train, itself a possible real-world factor), PGD-based radii for trained models, and, on real data, only an MLP on MNIST in a rotated basis (E6). The strongest next test is a trained deep network on a dataset where the number of dimensions carrying a fixed amount of evidence can be varied in the native basis.

A Reproducibility and relation to the earlier note

All experiments are in `root_cause_experiments.py`, E5 in `root_cause_factorial.py`, E6 in `root_cause_realdata.py`, E6b in `root_cause_cifar.py`, and all figures and numbers in `root_cause_figures.py`; E1–E5 take about seven minutes on CPU and E6 and E6b about ten minutes each on one GPU. Training for E1–E5: Adam, learning rate 0.02, 500 full-batch steps, weight decay 10^{-2} , 4,000 training and 2,000 test points, seeds 0–9 reseeding both data and initialisation. Minimal radii: bisection (14 iterations, search range $[0, 4]$) over 20-step PGD with step $\varepsilon/4$, counting a point as flipped if *any* iterate is misclassified; ℓ_2 attacks use normalised gradient steps with projection onto the ball.

What changed from the earlier note [1]. (a) Inputs are unclipped, which removes the $\varepsilon = 0.5$ artifact and makes the Bayes rule exactly linear. (b) The metric is the minimal flip radius, not success on a grid. (c) The earlier headline that “non-robust-only attacks are 2–4 $\times$ more effective than robust-only attacks” is replaced by E3, which shows the comparison reflects five attackable dimensions versus one. (d) The earlier claim that removing the weak dimensions makes the model “substantially harder to fool at every budget from $\varepsilon = 0.05$ ” was too strong: in that note’s own Table 2 the two models are no different below $\varepsilon = 0.10$ (B is slightly *worse* at 0.05). The gain appears at larger budgets, consistent with the median radius of 0.28 found here. (e) Material on activation sparsity, Model C initialisation, pruning, and seed-dependent collapse is not needed for the claim and is omitted.

References

- [1] Companion note: *Robust and Non-Robust Features in a Fully Controlled Synthetic Classification Task: A Minimal Demonstration* (`toy_robust_features_paper.tex`, this repository).
- [2] I. Goodfellow, J. Shlens, C. Szegedy. *Explaining and Harnessing Adversarial Examples*. ICLR, 2015.
- [3] A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, A. Madry. *Adversarial Examples Are Not Bugs, They Are Features*. NeurIPS, 2019.
- [4] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, A. Madry. *Robustness May Be at Odds with Accuracy*. ICLR, 2019.

- [5] L. Schmidt, S. Santurkar, D. Tsipras, K. Talwar, A. Madry. *Adversarially Robust Generalization Requires More Data*. NeurIPS, 2018.
- [6] C.-J. Simon-Gabriel, Y. Ollivier, L. Bottou, B. Schölkopf, D. Lopez-Paz. *First-order Adversarial Vulnerability of Neural Networks and Input Dimension*. ICML, 2019.